Screened nuclear-recoil transport in CYGNO-MC

Physics explanation and comparison with the standard branch

22 October 2026

Screening implementation: `g4v11.3_cygno04_screening`, source revision `92dd4e9`. Standard-branch reference: `g4v11.3_cygno04`, commit `3f53d24ce433e304dd86f31ef9c5f16cefd50ee1`. The retained simulations used **Geant4 11.4.0 and ROOT 6.36.08 under WSL**. The branch names are historical; these results do not establish a tested Geant4 11.3 build.

1. What the new branch is trying to describe

A nuclear recoil is an atom or ion moving through the detector gas after receiving kinetic energy. As it slows down, it interacts with both the electrons and the nuclei of surrounding atoms. Interactions with electrons produce excitation and ionisation. Encounters with nuclei transfer motion to other atoms and bend the recoil's trajectory. The balance between these two mechanisms determines how far the original ion travels, how much it turns, and how much of its initial energy becomes electronic excitation and ionisation.

The standard CYGNO branch already simulates electronic energy loss and nuclear interactions. The screening branch changes the low-energy nuclear transport for selected ions: energy transfer and deflection are sampled through the TestEm7 screened-collision process. Electronic stopping from Geant4's option4 electromagnetic physics remains active. The change is therefore more specific than replacing the entire electromagnetic physics list.

Two additional choices matter. First, an atom struck by a screened collision does not start a separately transported recoil track: its energy is deposited locally. Second, CYGNO's existing TRIM response function is integrated over each finite energy deposit, replacing the old pointwise weight. These choices affect different outputs. The first changes the spatial description; the second changes the custom electronic-response estimate. Keeping them separate is essential when interpreting agreement with TRIM.

The retained comparison finds substantially improved average endpoint-range agreement for H, C, O and F. He endpoint agreement is almost unchanged. Endpoint-direction agreement improves for all five ions. The custom QF also follows the imported TRIM response closely, but that agreement follows from the response prescription and is not an independent prediction of microscopic ionisation.

Terms used below

Term	Plain-language meaning
Primary ion	The ion placed in the gas at the start of an event.
Target recoil	An atom set in motion when the primary or another moving atom strikes it.
Electronic stopping	Energy lost through interactions with electrons, including excitation and ionisation.
Nuclear stopping	Energy transferred to atomic motion through encounters with nuclei.
Screening	Reduction of the interaction between two nuclei by the surrounding electrons.
Transport step	A short segment over which Geant4 advances a particle and applies the active processes.
Local deposit	Energy assigned to the current step rather than carried away by a new tracked particle.
NIEL	Non-ionising energy loss, the part of the local energy deposit classified as non-ionising by the physics model.
QF	Here, the imported TRIM mean electronic-energy fraction; it is not a measured collected-charge fraction.

2. The same event in the two branches

Both branches start from the same kind of source: one H-1, He-4, C-12, O-16 or F-19 primary in He/CF4 gas. The production macros sample total ion energy from 0.25 to 1000 keV, uniformly in its logarithm. Equal logarithmic intervals therefore receive equal probability; a factor-of-ten interval at low energy has the same probability as a factor-of-ten interval at high energy. Initial directions are isotropic, meaning that all directions in space are equally likely.

The source position is the active-gas centre at (0, 0, 245) mm. Gas conditions are 60/40 He/CF4 by molecular fraction, 900 mbar and 293.15 K. The comparison keeps geometry, gas composition, source distributions and production cuts fixed. The screened process and step limits differ; the meaning of an energy in keV remains **total energy per ion**, not energy per nucleon.

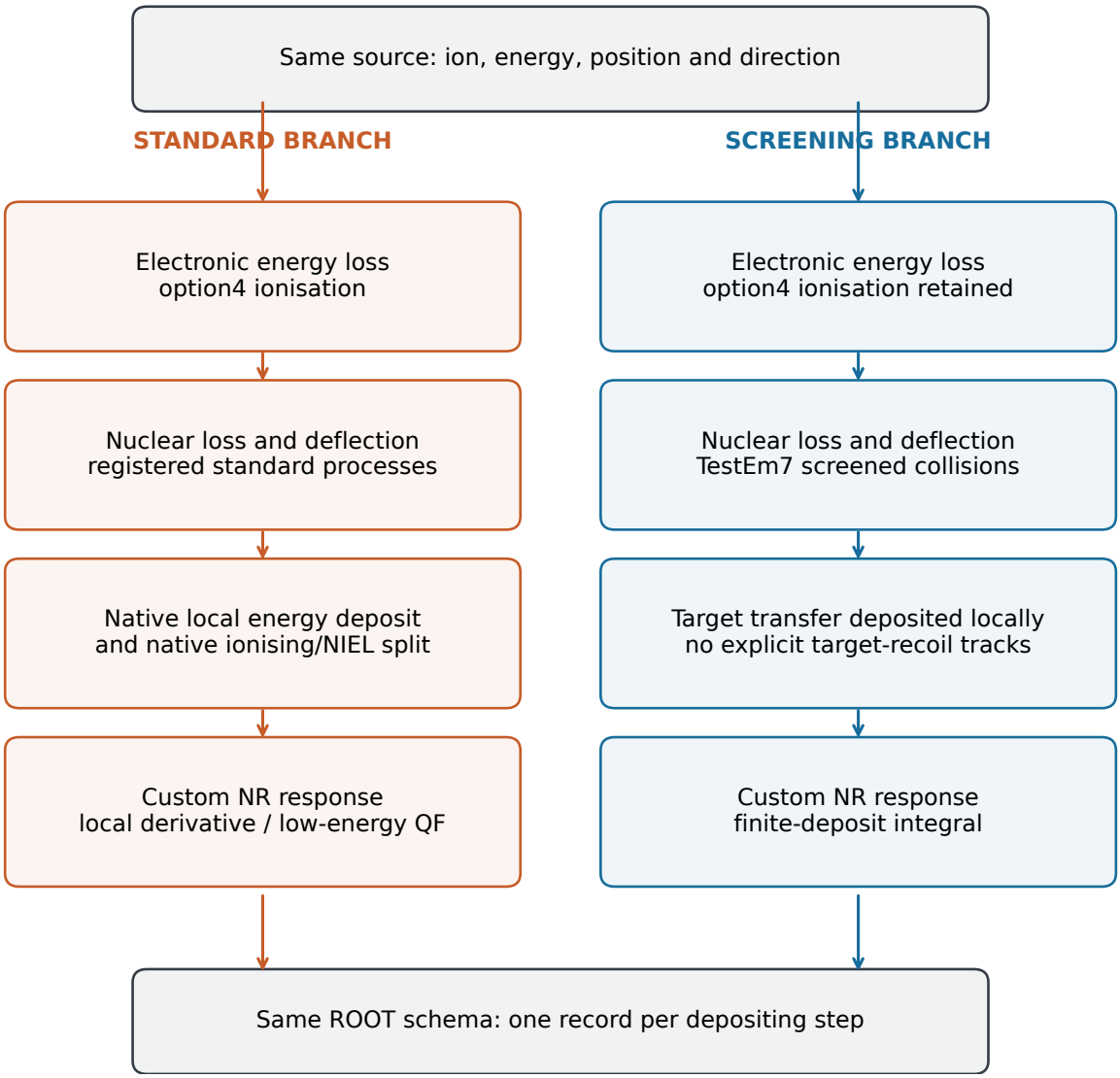


Figure 1: Transport and output flow in the two branches. Boxes show the implemented sequence, not simulated trajectories. Electronic stopping and the ROOT output structure are shared; nuclear transport and custom response weighting differ.

In the standard branch, the registered electromagnetic processes provide nuclear stopping and ion scattering alongside electronic stopping. Nuclear energy loss and angular deflection are handled through those processes and their models. The event's local deposits are then weighted with the existing custom SRIM response.

In the screening branch, the standard processes are constructed first. The code then removes nuclear stopping and multiple/Coulomb scattering from the selected ion process managers and installs the screened-collision process. Electronic ionisation processes remain. A high-energy multiple scattering model is also registered, but its activation threshold is far above the recoil energies tested here.

This comparison must not be read as "the old branch had no screening". Standard Geant4 scattering models also account for atomic physics. The specific difference is the replacement of the selected nuclear-loss and scattering processes by the imported TestEm7 ZBL collision treatment, together with the deposition, resolution and response choices explained below.

Part of an event	Standard branch	Screening branch
Initial ion and gas	Same source and material setup	Same source and material setup
Electronic stopping	option4 ionisation processes	Retained
Low-energy nuclear transport	Standard registered nuclear stopping and scattering	TestEm7 screened elastic collisions
Target energy from TestEm7 collisions	TestEm7 process absent	Deposited locally, without an explicit recoil cascade
Custom response per NR hit	Local deposit times derivative; QF multiplier at or below 25 eV	Integral of the same response over the finite deposit
Spatial step limits	Original species-dependent caps	Larger caps, with the same energy dependence
Saved event format	Native depositing steps	Same schema and hit-creation rule

3. What a screened collision means physically

Why the force becomes weaker at larger separation

Two positively charged nuclei repel one another. If they were bare nuclei, the repulsion would follow the Coulomb potential. Real atoms also contain electrons. At larger separations the electrons partly shield the nuclear charges, so the interaction is weaker than that between bare nuclei. This is the screening represented by the Ziegler-Biersack-Littmark, or ZBL, potential selected by the branch.

The imported code writes the potential as a Coulomb interaction multiplied by a screening function. The function is close to one at very small separation and decreases as separation grows. The screening length sets the distance scale and depends on the charges of the two nuclei. No ZBL coefficients or screening lengths are fitted to the retained CYGNO comparison.

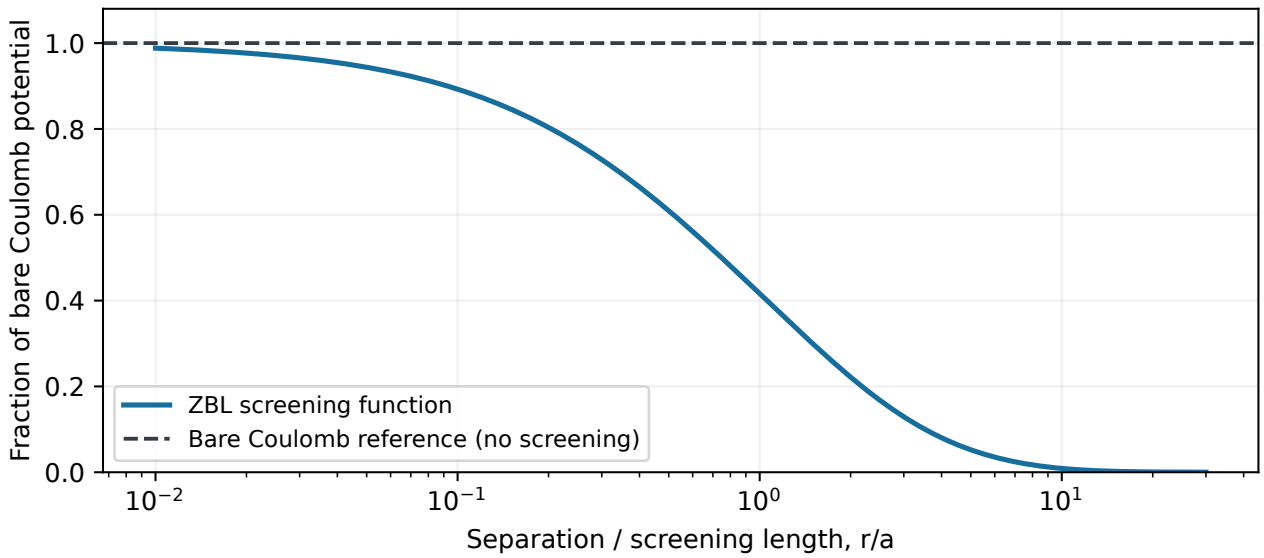


Figure 2: The ZBL screening function used by the imported code. The horizontal reference represents bare Coulomb repulsion, not the standard CYGNO branch. The curve is an analytic illustration of the implemented potential, not a measured scattering distribution.

A close encounter can produce a relatively large transfer and bend the primary strongly. A distant encounter generally produces a smaller effect. The outcome also depends on the masses and incident energy: the same geometric encounter need not produce the same transferred energy for H and F. A primary's final displacement emerges from many such encounters combined with electronic energy loss along its route.

This coupling between nuclear energy transfer and deflection is the physical motivation for the replacement. It offers a way to change the range and bending together. The comparison below tests the outcome of the complete implementation; it does not isolate the potential choice from all the other changes in the branch.

How Geant4 advances the ion

The simulation does not march the ion forward using one fixed distance. Several processes propose distances to the next interaction or limit the step length. Geometry boundaries also limit how far a step can go. Geant4 uses the competing constraints to advance the track. Along a step, electronic processes can remove energy; at a screened nuclear interaction, the collision process can change both energy and direction.

The screened process first constructs cross-section tables. A cross section describes the effective probability of an interaction with a target species. Combined with the number of atoms per unit volume, it sets a mean free path: the characteristic distance between proposed interactions. In a gas mixture the inverse mean free path is built from the sum of number density times cross section for each element. These tables are computed and cached for reuse, rather than rebuilt after every step.

At a proposed collision the code samples an impact parameter, the sideways offset of the encounter, and selects a target nucleus. The upstream target selector samples element abundance by the number of atoms per unit volume; isotope selection follows the element's abundance information. Some proposed encounters are rejected as virtual collisions and leave the primary unchanged. For an accepted encounter the classical screened kinematics calculates the energy transfer and scattering geometry. The projectile's momentum is updated consistently with that sampled recoil momentum.

Disabling target-recoil tracks does **not** make the primary continue straight. The collision still subtracts energy from the primary and changes its direction. What is omitted is the later motion of the struck atom.

4. Where the transferred energy goes

A tracked recoil and a local deposit describe different spatial models

In a fully transported cascade, a struck atom receives kinetic energy, travels some distance, collides with more atoms, and creates further branches. That cascade can spread the electronic energy and atomic motion over a volume around the primary track.

The implemented screening configuration instead places the target's transferred energy at the collision step. This reduces the number of particle tracks and depositing steps that must be written. It retains primary deflection and energy loss but omits the target-recoil branch's subsequent displacement and correlations. It is a deliberate approximation to spatial transport, rather than an output-file encoding technique.

As a simple accounting example, suppose an ion has 10 keV just before a nuclear encounter and transfers 2 keV to a target. Ignoring simultaneous electronic loss for the example, the primary continues with 8 keV. The other 2 keV is recorded locally. In a tracked cascade it would initially be kinetic energy of a new target-ion track; in the present configuration there is no such moving secondary. The example illustrates accounting only; it is not a sampled event from the retained comparison.

When the residual projectile energy becomes smaller than the process's terminal cutoff, that remainder is also deposited and the projectile energy is set to zero. This prevents the terminal kinetic energy from being silently discarded. Other active processes can still create secondary electrons or other particles; the recoil-generation switch is specific to the screened process.

Three energy quantities must be kept separate

The screened process divides its local target deposit into electronic and non-ionising parts using the native Lindhard-Robinson partition. If its non-ionising fraction is `p`, a transfer `T` contributes `T*p` to NIEL and `T*(1-p)` to native electronic deposition. Their sum is the raw local deposit `T`.

The partition selects the material element with the largest number of atoms per unit volume as its effective target. In 60/40 molecular He/CF4, relative atomic counts are 0.6 He, 0.4 C and 1.6 F, so fluorine supplies that effective target. This is distinct from simulating a full mixture-specific recoil cascade. It is also distinct from CYGNO's imported TRIM response.

Energy quantity	How it is obtained	What it can tell us
Raw deposited energy	Sum of native local deposits	Whether deposited energy is compatible with the initial energy budget
Native Geant4 ionising energy	Total local deposit minus native NIEL	Electronic deposition according to the active Geant4 processes and partition
Custom TRIM response	Apply the imported response function to NR deposits	The prescribed mean electronic-response estimate used by CYGNO

Changing the custom response does not repair raw-energy non-conservation: raw deposition is saved before weighting. Likewise, agreement of the custom response with TRIM does not demonstrate that the native Geant4 ionising channel has independently reproduced TRIM.

5. How finely the track is simulated

Two controls influence how many steps are needed. They operate at different levels and must not be confused.

The **transfer-resolution parameter** enters the construction of screened collision cross sections. The branch uses `0.001*E`, bounded below by 0.1 eV and above by 100 eV, where `E` is total projectile kinetic energy. At low energy the parameter is small, allowing finer treatment of soft encounters. At higher energy it becomes larger and reduces the number of native collision steps. It is a parameter of the approximate table and sampling construction, not an exact guarantee that every accepted collision transfers more energy than that value.

The **maximum spatial step** is an upper limit on how far the ion may advance in one step. It scales as the macro's cap multiplied by `min(1, sqrt(E/40 keV))`. Below 40 keV the permitted step becomes shorter as the ion slows; above 40 keV it reaches the cap. Actual steps can be shorter because collisions, electronic stopping and geometry compete with this limit. Increasing a cap permits longer native steps; it does not merge hits after transport.

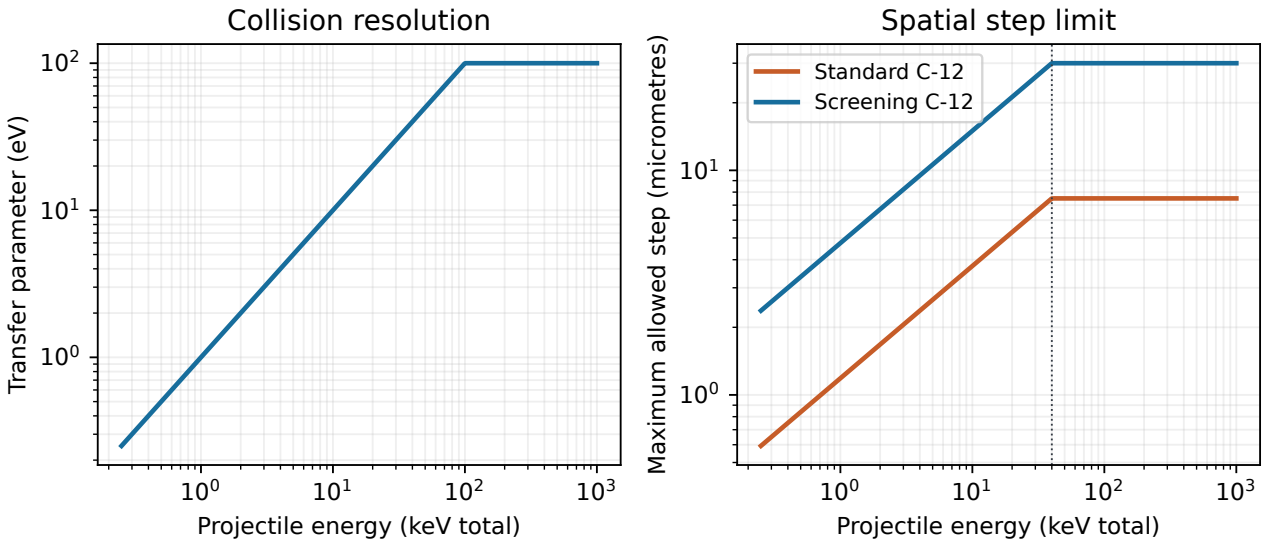


Figure 3: Two independent resolution controls. Left: the transfer parameter used in cross-section construction. Right: the maximum spatial step for C-12 in each branch. These curves are calculated from the configured laws, not measured mean step lengths.

Ion	Standard cap (micrometres)	Screening cap (micrometres)	Increase
H-1	52	208	4 times
He-4	26	208	8 times
C-12	7.5	30	4 times
O-16	6.5	26	4 times
F-19	6.2	24.8	4 times

For example, at 1 keV the maximum C-12 step is approximately 1.19 micrometres in the standard branch and 4.74 micrometres in the screening branch. Both values are upper bounds. With a coarser allowed step, a pointwise response weight becomes more sensitive to how the energy loss is divided between steps. This is one reason finite-deposit integration matters in the new configuration.

Three low-energy settings also have different jobs. The screened process's constructor supplies a 0.1 eV physics/table limit and an approximately 1 eV-per-nucleon terminal recoil cutoff. The macros separately keep `lowestMuHadEnergy` at 25 eV for the electromagnetic treatment of charged hadrons/ions. These are different controls, not three alternative values for a single threshold. The process receiving a track and the competing transport mechanisms determine its terminal handling.

Transport accuracy cannot be inferred from a small ROOT file. Varying the transfer parameter and the spatial caps is needed to establish convergence: the relevant ranges, bending and energy budgets should approach stable values as resolution is increased. The retained comparison has not established that convergence.

6. Why the custom response calculation changes

The response curve stays the same

For each supported ion, the existing header contains a TRIM-derived mean electronic-energy fraction `QF(E)`. Multiplying that fraction by ion energy gives the mean response `R(E) = E*QF(E)`. The coefficients are shared by the two reference branches. The fit covers 25 eV to 3 MeV total ion energy; below 25 eV the existing smoothly matched power-law continuation is used. The retained transport comparison covers the narrower 0.25-1000 keV initial-energy interval.

TRIM electronic energy includes excitation. Neither the fit nor its integration converts that energy into experimentally calibrated collected charge or light. The source low-energy extension also has unresolved energy-closure failures: all 45 new 25/40/63 eV batches failed the source's 1% criterion, with a maximum discrepancy of 14.58%. Those nominal curves are retained with that limitation; improved Geant4 transport does not remove it.

The standard weight uses the slope at one point

Above 25 eV, the standard branch estimates a hit's custom response as raw local deposit multiplied by the derivative `dR/dE` evaluated at the pre-step energy. A derivative is the local slope of a curve. Multiplying one slope by an energy interval approximates the change across that interval. It is accurate for a sufficiently small interval, but a curved response can change slope within a large deposit.

At or below 25 eV, the standard event action calls a second method which multiplies by QF instead. This is the dispatch at the frozen base commit; the older 1 keV switch in historical pilot material is not the rule in that base. In the screening branch, both methods now call the same finite-deposit calculation.

The new weight follows the curve across the deposited interval

For pre-step energy `E` and raw local deposit `d`, the new hit response is:

```
response of the hit = R(E) - R(max(0, E-d))
```

Rather than extending the tangent from one endpoint, this takes the change in the response curve over the whole interval. For a terminal local deposit with `d = E`, it returns `R(E)` because `R(0) = 0`. For extremely small relative deposits, below one millionth of `E`, the code uses the derivative at the interval midpoint to avoid subtracting nearly equal numbers. That is a numerical approximation for stability; it does not discard a hit.

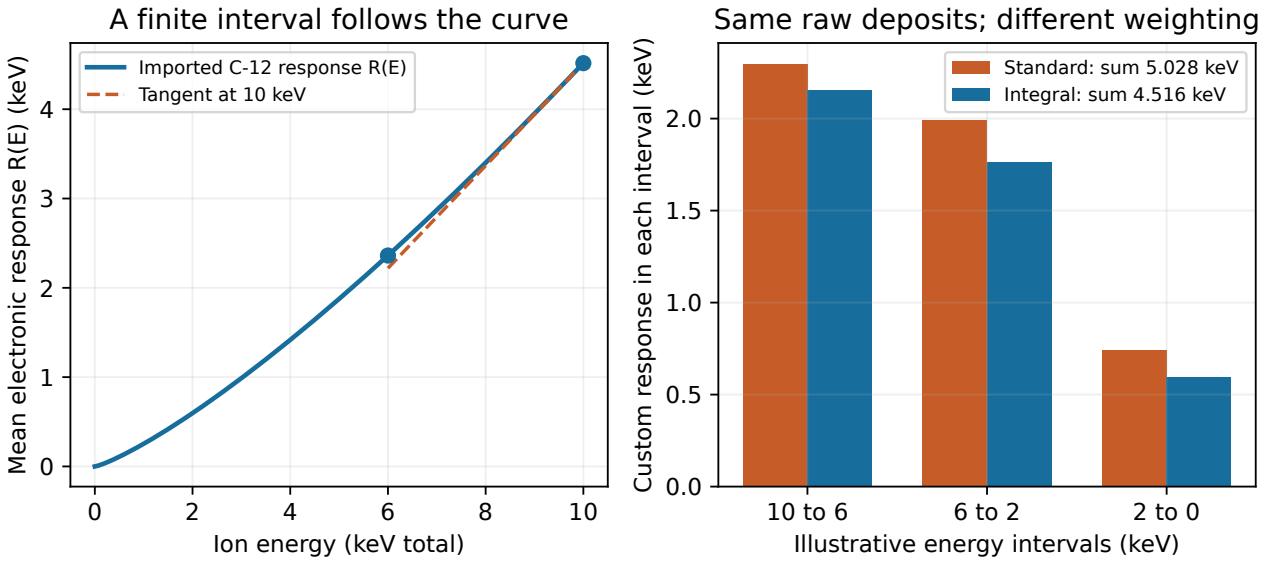


Figure 4: Finite-deposit response weighting illustrated with the actual imported C-12 response. The left panel contrasts the curved response with its tangent at 10 keV. The right panel applies both prescriptions to the same three prescribed local energy intervals. The intervals are illustrative, not a simulated track.

The following example makes the difference explicit. A hypothetical C-12 primary loses its energy locally in three intervals: 10 to 6, 6 to 2, and 2 to 0 keV. Every interval begins above 25 eV, so the standard column uses its derivative rule. Values come from a small C++ evaluator compiled against the actual response header, without running transport.

Illustrative step	Energy before / after (keV)	Raw deposit (keV)	Standard custom response (keV)	Integrated custom response (keV)
1	10 / 6	4	2.2960	2.1541
2	6 / 2	4	1.9917	1.7652
3	2 / 0	2	0.7404	0.5965
Total	10 / 0	10	5.0281	4.5157

Entries are rounded to four decimals; the total uses unrounded values.

The integrated responses add to `R(10 keV)`: the response at each internal boundary appears once with a plus sign and once with a minus sign, and cancels. Subdividing the same ideal local-loss sequence therefore preserves the total, apart from the tiny-deposit numerical approximation. The point-derivative estimate need not have that property.

This cancellation also explains the very close custom-QF agreement in the retained screening results. For a contained primary with entirely local losses, the prescription enforces the imported mean response. It does not show that screened collisions independently predicted that QF. If a tracked secondary carries energy away, `E-d` is not necessarily the actual post-step primary energy; the cancellation need not hold. Unsupported PDGs retain raw deposition. No new microscopic fluctuation or Fano-statistics model is added.

7. From a physical step to the saved ROOT event

The sensitive detector reads the total deposit and the native NIEL for each step in sensitive gas. Optical photons are excluded, and a step with zero total deposit creates no hit. Each other depositing step creates one hit containing the raw energy, native ionising energy, pre-step kinetic energy, position, step length, time and particle/track identifiers. Energies are saved in keV, lengths in mm and times in seconds; the saved hit position is the post-step position.

The event action saves raw and native ionising values before applying the custom response to nuclear-recoil hits. Protons and ions are included in that classification. It then fills aligned hit vectors and sums the corresponding event energies. Non-NR hits contribute raw energy to the total-response channels. Neither the writer nor the sensitive detector aggregates, thins or quantises the depositing steps.

Event energy branch	Contents
`energyDep`	Raw deposited energy from all accepted hits
`energyDep_NR`	Raw deposited energy from proton/ion hits
`energyDep_NRQF`	Custom TRIM response from proton/ion hits
`energyDep_QF`	Custom NR response plus raw non-NR deposition
`energyDep_NRQF_geant`	Native Geant4 ionising energy from proton/ion hits
`energyDep_QF_geant`	Native Geant4 NR ionising energy plus raw non-NR deposition

The equivalent six hit-energy vectors retain the same schema and sum to those event scalars. `numhits` still counts native depositing steps. The initial energy in `tree1/Energy` is in MeV and must be multiplied by 1000 when compared with these keV quantities.

An unchanged file layout does not imply identical spatial physics. With local target deposition, the target's energy is attached to the transporting projectile's step and PDG. There is no target-ion track whose later positions could be recovered by a different ROOT reader. This distinction matters for charge-cloud shape and head-tail studies.

8. How the comparison with TRIM is constructed

The retained comparison contains **10,000 events per ion per branch**, 100,000 events in total. Source arrays match bitwise between branches: the energies, starting positions, directions and IDs are the same. The subsequent collision histories are not matched; changing the physics changes both interactions and random-number consumption. The standard control was built from an immutable archive of the base commit. All retained primary tracks stop inside sensitive gas.

The TRIM reference contains 195 energy/species points from 897 batches and 89,700 simulated ions. These are simulated references, not detector measurements. The analysis interpolates the reference to each Geant4 primary's initial energy. Events are grouped into ten equal logarithmic energy bins. Plotted energy coordinates are geometric means of the energies in each bin.

Before looking at the plots, the observables need precise definitions:

Observable	Definition	How to interpret it
Endpoint range	Straight-line distance from the source to the final primary position	How far the primary ends from its starting point
Projected range	Endpoint displacement projected along the initial direction	How far forward it ends; an individual event can be negative
Endpoint cosine	Projected range divided by endpoint range	1 is straight forward, 0 is transverse, negative is backward
True track length	Length accumulated along the curved trajectory	Different from endpoint range; not equated with a sampled TRIM polyline

The ratio plots use the mean of **individual event-to-reference ratios**, not a ratio of two bin means. A value of one means agreement with the interpolated TRIM expectation. A value of 0.7 means a mean relative range 30% below that reference. The plotted ratio uncertainty combines finite Geant4 statistics with the propagated conditional TRIM mean uncertainty. It excludes interpolation and model systematic uncertainties. Direction plots show Geant4 statistical standard errors; no TRIM uncertainty band is supplied there. Connecting lines guide the eye between bins.

Endpoint range: a large improvement, with a remaining energy dependence

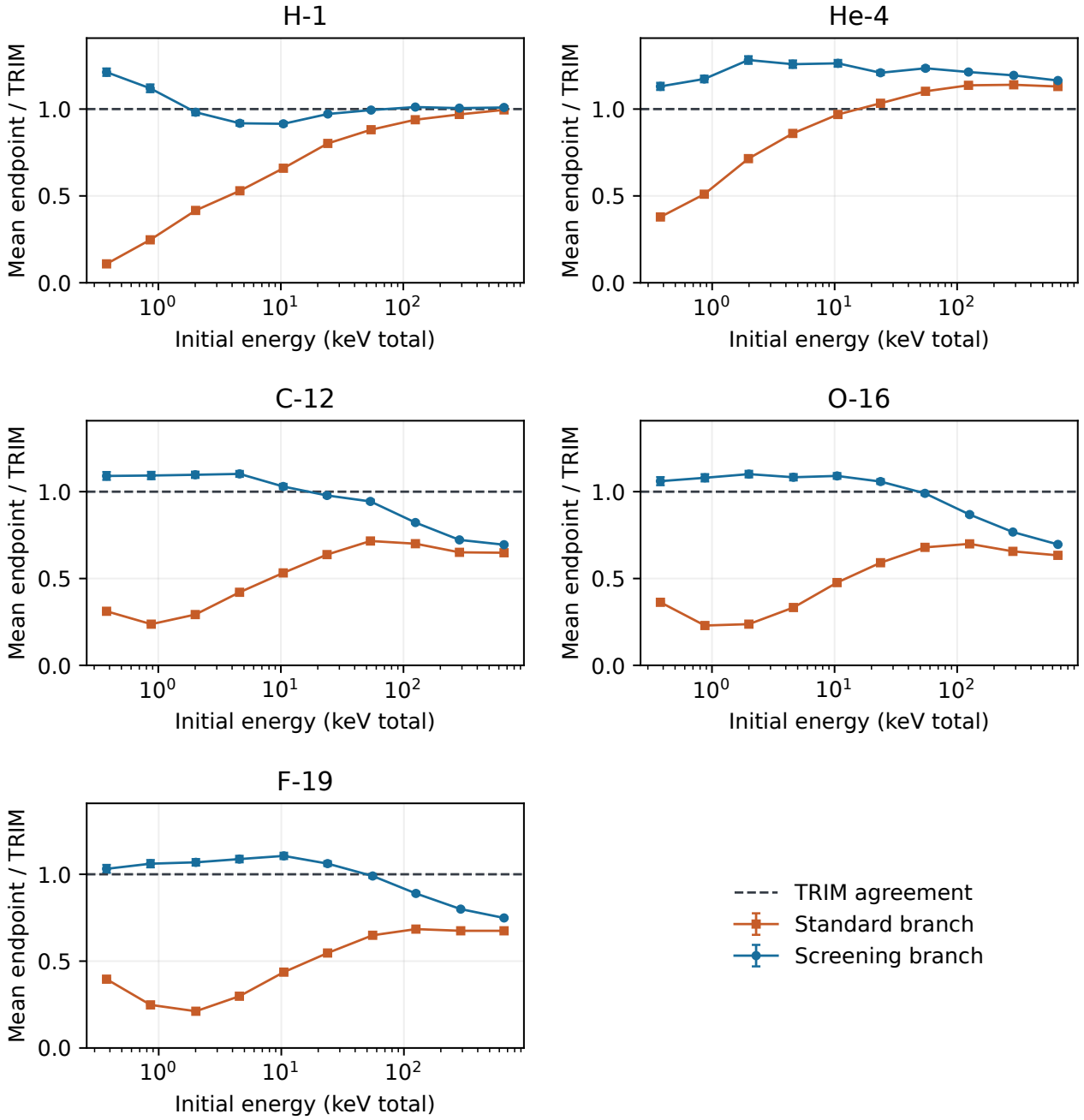


Figure 5: Endpoint-range ratios to TRIM versus initial energy. Each point summarises one logarithmic energy bin, with the retained combined statistical uncertainty. The dashed line marks agreement. The same vertical scale is used in all species panels.

For H, C, O and F the screening curves are generally closer to one than the standard curves, consistent with the smaller average discrepancies in the table below. He does not show the same endpoint improvement. The benefit is also not uniform across energy: in the highest bin near 0.66 MeV, C and O endpoints remain around 30% shorter than TRIM and F around 25% shorter. These residuals remain visible even when an overall average score looks favourable.

The outcome is physically plausible because a different combination of nuclear energy transfer and deflection changes the accumulated slowing and final displacement. It cannot be assigned to the ZBL potential alone: local recoil deposition, the adaptive collision resolution and the larger spatial caps are part of the tested configuration.

Projected range: forward progress combines length and bending

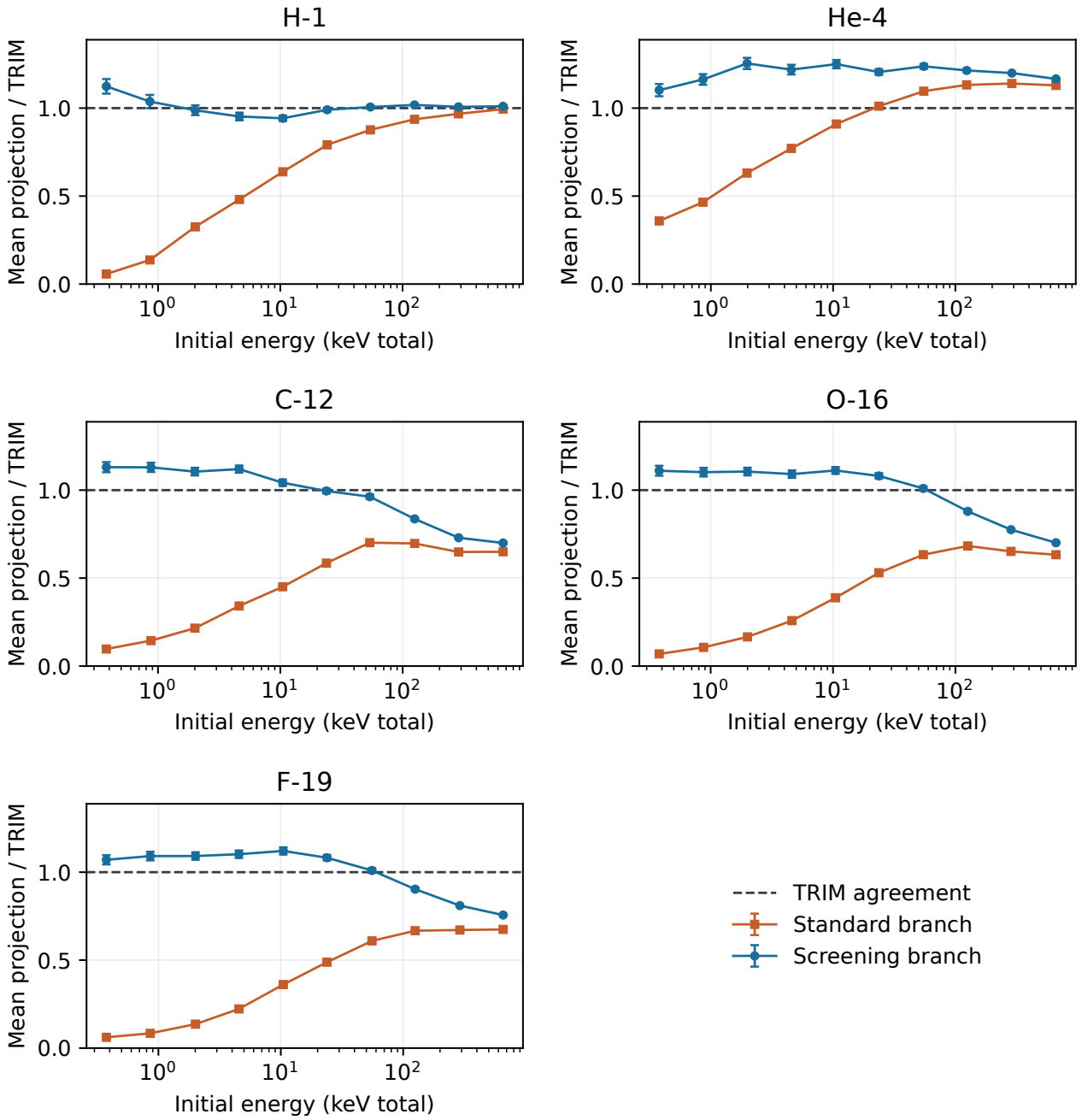


Figure 6: Projected-range ratios to TRIM. These compare final forward displacement rather than total curved path length. Agreement can change through both the stopping distance and the amount of bending.

A particle can have a reasonable endpoint distance and still make too little forward progress if it bends excessively. Projected range therefore adds information beyond the endpoint magnitude. The screening branch improves the average projected-range discrepancy for all five ions in the retained comparison, although He still has a discrepancy near 20% and high-energy C/O/F residuals remain.

Endpoint direction: improved primary displacement is not detector resolution

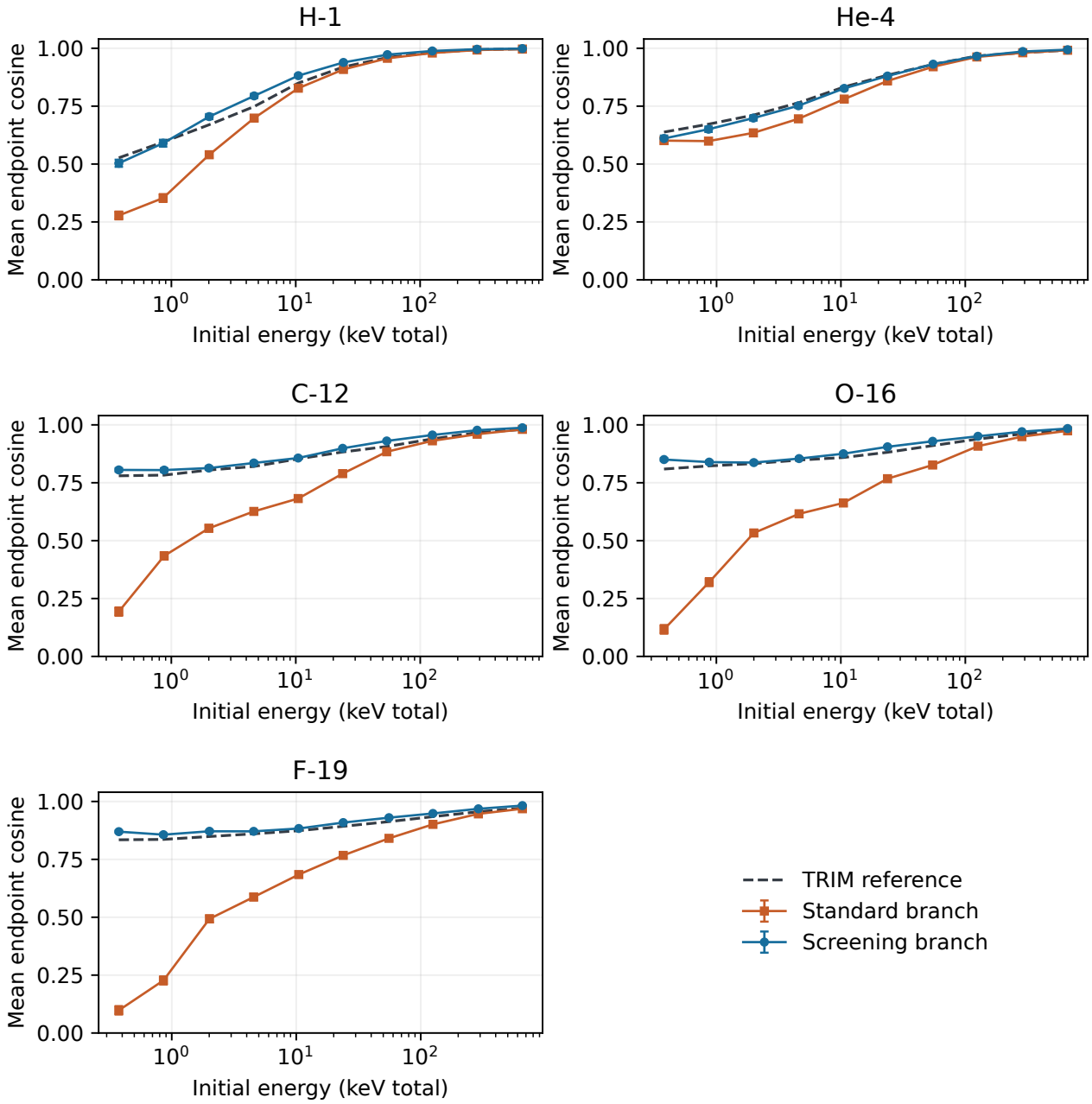


Figure 7: Mean endpoint cosine compared with the TRIM reference. Error bars are Geant4 statistical standard errors. A larger cosine means the final displacement points more strongly along the initial direction. These are primary endpoint observables, not final-momentum or reconstructed-angle measurements.

The screening branch follows the reference mean endpoint cosine more closely for all five species in the bin-averaged score. This supports improved primary bending behaviour for this source and gas. It does not establish a detector angular resolution: reconstruction also depends on the spatial distribution of ionisation, diffusion, amplification, readout and analysis. The omitted target-recoil branches are particularly relevant to that spatial question.

Numerical overview of the range and direction comparison

Ion	Endpoint discrepancy, standard / screening (%)	Projected discrepancy, standard / screening (%)	Cosine discrepancy, standard / screening
H-1	34.52 / 5.77	37.96 / 3.30	0.0711 / 0.0183
He-4	21.10 / 21.23	23.73 / 20.10	0.0349 / 0.0094
C-12	48.52 / 12.52	54.69 / 13.04	0.1684 / 0.0149
O-16	51.00 / 11.51	58.81 / 12.56	0.2165 / 0.0156
F-19	51.80 / 9.88	60.24 / 10.99	0.2407 / 0.0169

For range, the score is the average over ten bins of the absolute percentage difference from a ratio of one. For cosine, it is the average absolute difference from the reference mean cosine, without a percentage normalisation. Every bin has equal weight. These are descriptive scores, not significance-weighted goodness-of-fit tests; they do not imply agreement at every energy or quantify the probability that a model is correct.

9. Energy conservation and output size

Raw energy is checked independently of custom quenching

For a contained primary, the raw energy deposited in sensitive gas should be compatible with the initial energy after accounting for energy carried elsewhere. An excess above the initial energy cannot be explained by a change in QF, because QF is applied after raw deposition is saved. The retained check counts events whose raw deposit exceeds the initial energy by more than 0.01%, that is, a ratio greater than 1.0001.

Ion	Standard events above the tolerance / 10,000	Screening events above the tolerance / 10,000	Largest standard excess (%)
H-1	2	0	0.016
He-4	50	0	2.356
C-12	66	0	5.626
O-16	63	0	7.330
F-19	54	0	8.119

There are 235 such events among 50,000 standard-branch events and zero among 50,000 screening events. No raw-energy rescaling is used. This is useful evidence for the retained configuration, but zero observed failures in a finite sample is not a proof of energy conservation in every possible run. The comparison changes multiple transport choices together; it does not isolate the cause of the control's excesses.

Smaller files result from fewer native deposits

Local target deposition avoids explicit secondary-recoil branches, and the transport settings limit the number of native steps. The resulting files still contain one record per native depositing step. They do not require a new reader or a compact encoding.

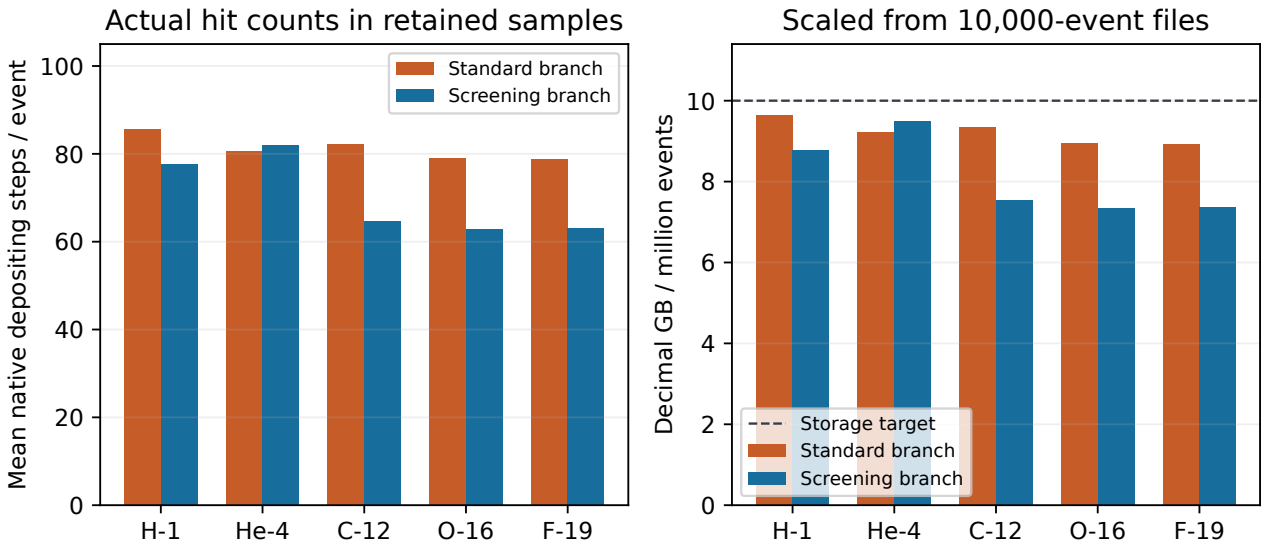


Figure 8: Native hit counts and storage in the retained samples. Hit counts are measured means. File sizes are extrapolated from actual 10,000-event files to one million events; the dashed line is the 10 decimal GB target.

Ion	Standard GB per million	Screening GB per million	Screening mean hits per event
H-1	9.638	8.786	77.6
He-4	9.217	9.494	82.0
C-12	9.335	7.545	64.7
O-16	8.955	7.335	62.8
F-19	8.923	7.354	63.0

All screening projections are below 10 decimal GB per ion per million events for the stated log-uniform source. He has only about 5% margin, and its file is slightly larger than the standard control's. These are sample-size extrapolations, not measured million-event files. A sample concentrated at high energy, different step caps or different geometry needs another size assessment. Hit counts alone are not a wall-time benchmark.

10. What the retained evidence supports

The source comparison and retained checks support a concrete conclusion: the tested screening configuration improves several primary transport observables while preserving the existing ROOT schema and staying within the pilot storage target. That conclusion has a defined scope, shown below.

Question	Evidence available	Remaining limitation
Are primary ranges closer to TRIM?	Improved H/C/O/F endpoint scores and all five projected-range scores	He endpoint unchanged; high-energy C/O/F residuals
Is primary endpoint bending closer?	Improved mean-cosine scores for all five ions	Endpoint displacement is not a reconstructed detector angle
Is the raw budget improved?	No screening events above the excess tolerance in 50,000 events	Finite sample; step/cutoff convergence not established
Does custom QF follow TRIM?	Very close agreement with the imported response	Largely imposed by finite-deposit integration
Is the output compatible?	Branch names/types, aligned vectors and all six energy sums checked	Same schema does not restore omitted target tracks
Is a full charge cloud validated?	Primary transport and model-to-model spatial diagnostics	No per-event TRIM 3D charge-cloud reference; recoil branches omitted
Is charge or light yield calibrated?	Nominal electronic-energy response	Experimental response and intrinsic fluctuations unresolved

The next physical qualifications are resolution-convergence studies and an assessment of the omitted recoil-cascade spatial structure. The SRIM source's low-energy closure and threshold uncertainties also remain. TRIM is a useful model reference, but agreement with it is not a substitute for detector measurements. A more complete detector prediction needs charge/light conversion, transport of the signal, readout response and reconstruction on top of the present primary-ion transport.

Appendix A. Mapping the explanation to the implementation

Process installation and affected particles

`CYGNOPhysicsList::ConstructProcess()` constructs the registered modular physics and then installs:

```
auto* screened = new G4ScreenedNuclearRecoil(
    "ScreenedElastic", "zbl", false, 1.*eV, 0.1*eV);
screened->SetMaxEnergyForScattering(100.*MeV);
```

`"zbl"` chooses the screened potential. `false` disables explicit target recoil generation. The final two energies set the mass-scaled terminal recoil cutoff and initial physics/table limit. The replacement loop selects `GenericIon`, `alpha`, `He3`, `proton`, `deuteron` and `triton`. GenericIon provides the setup for dynamically created heavy ions, including the C/O/F primaries.

The code removes `nuclearStopping` by name and multiple/Coulomb scattering by process subtype, iterating backwards through the process list. A failed registration raises `ScreeningRegistration` as a fatal exception. Option4 electronic processes, decay and the existing hadronic constructors remain registered. Electron/photon process managers are not replaced.

A new `G4hMultipleScattering` contains an Urban model activated at 100 MeV total. The screened process's maximum is interpreted approximately per nucleon. These are different energy conventions and do not establish a validated high-energy handoff. The retained NR results stop at 1 MeV total, far below those boundaries; arbitrary high-energy-ion use is not qualified by this comparison.

Collision and energy-accounting entry points

Source function	Role in the event
`GetMeanFreePath()`	Creates/caches projectile-charge handlers and interpolates material interaction lengths
`BuildMFPTables()`	Combines elemental cross sections and atomic number densities
`PostStepDoIt()`	Handles the terminal remainder or proposes and samples a collision
`DoScreeningComputation()`	Solves the screened scattering geometry
Classical `DoCollisionStep()`	Calculates transferred recoil energy and chooses local deposition or recoil creation
`G4SingleScatter::DoCollisionStep()`	Updates projectile momentum/energy and creates a target secondary only when enabled
`DepositEnergy()`	Divides local energy into native electronic deposition and NIEL

The classical transfer is proportional to `4*E*a1*A*cos(zeta)^2/(a1+A)^2`, using the effective masses and angle calculated by the imported kinematics. The local-deposition branch clears the recoil-secondary pointer and deposits its kinetic energy. When the primary's remainder is below the terminal cutoff, the code deposits that remainder too and proposes zero outgoing primary energy.

Three changes inside the imported TestEm7 process

The dependency is vendored from Geant4 v11.4.0 TestEm7 with its license and original notices retained. The application links two implementation units, the screened process and the Lindhard partition, plus the required interpolation headers. The complete TestEm7 detector and action classes are not imported.

The **adaptive transfer parameter** changes cross-section construction:

```
const G4double transfer =
    std::min(100.*eV, std::max(st.emin, 1.e-3*ee));
G4double eratio = (cmfact0 * transfer) / ee;
```

Here `ee` is total kinetic energy and `st.emin` receives the 0.1 eV physics cutoff. This is a transport-resolution approximation and needs physical convergence checks.

The **terminal guard** in `PostStepDoIt()` checks both the physics limit and the mass-scaled recoil cutoff, matching the condition in `GetMeanFreePath()`. It routes a terminal particle to local deposition before accessing collision handlers that may not have been initialised.

The **analytic ZBL tail** continues the same function beyond the finite screening table during scattering quadrature:

```
phi(x) = 0.1818 exp(-3.2x) + 0.5099 exp(-0.9432x)
       + 0.2802 exp(-0.4028x) + 0.0281 exp(-0.2016x)
```

Here `x` is separation divided by screening length. In-domain interpolation and the ZBL coefficients/length remain unchanged. The tail is specific to ZBL and must not be assumed to implement other potential keys correctly. The guard and continuation address numerical boundary handling; they are distinct from the transport-resolution approximation.

Custom response and saved-hit handling

`CYGNOHit::ApplyQuenching()` integrates the existing response over a finite raw deposit, using the midpoint derivative for very small relative deposits. `ApplyQuenchingAvg()` delegates to it. The model lookup, coefficients, energy-domain checks and sub-25 eV continuation remain in the existing response header/helper. Supported-ion energies above 3 MeV remain outside the response domain and cause an explicit error.

`CYGNOSensitiveDetector::ProcessHits()` still creates one hit for each accepted depositing step. `CYGNOEventAction` still classifies hits, fills vectors and forms the six energy sums. `CYGNORunAction` retains the ROOT schema. `CYGNOStepMax` retains the same energy-dependent cap law; only five NR macro cap values change. LF/CRLF differences should be excluded when comparing the working tree's otherwise unchanged source files.

Appendix B. Data, figures and reproduction

The source-level navigation is in the code walkthrough. The SRIM response note records the fitted response and historical import checks. Numerical transport results are retained in screening-comparison.csv, with hashes and sample checks in screening-validation.json. The existing comparison report contains additional observables.

All empirical curves in the figures come from the retained CSV or the retained score metadata. The flow diagram, potential and resolution curves are explanatory illustrations of the implementation. The response example uses the compiled source header on prescribed energy intervals. None of these figures introduces new Geant4 transport samples or new TRIM measurements.

To regenerate figures and the PDF from the repository root:

```
python3 docs/build_screening_note.py
```

The builder requires Python with NumPy/Matplotlib, g++, Pandoc and pdfLaTeX. It saves vector PDF and PNG figures under `docs/figures/screening/`, records their input provenance, compiles only the small response evaluator in a temporary directory, and refreshes the document hashes. The vector versions are embedded in the technical PDF.

To build the transport application separately:

```
cmake -S . -B build_screening -DWITH_GEANT4_UIVIS=OFF
cmake --build build_screening --parallel 3
```

Switching source branches requires a corresponding rebuilt or separate executable. The application uses the existing ROOT reader and NR macros. Dependency provenance and licensing are retained in external/TestEm7.